Constructing effective match strategies for the game of Tavli

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ABSTRACT

This paper presents a novel approach to constructing AI strategies for Tavli, a popular backgammon variant in Greece, comprising the games Portes, Plakoto, and Fevga. We develop Match Equity Tables (METs) specific to each game type in Tavli to estimate the probability of the first player winning under various match situations. Our methodology involves calculating these probabilities and utilizing them to create effective AI strategies tailored for Tavli matches. We evaluated the performance of our AI strategies against existing AI approaches and human players. The results show that our Tavli-specific AI strategy significantly outperforms humans and achieves a small but statistically significant improvement over other AI strategies. Furthermore, we examine the order of game types in a Tavli match and find that the traditional order (Portes-Plakoto-Fevga) offers the fairest advantage to the second player. Our findings indicate the potential for applying these methods to other match-based games and competition formats.

CCS CONCEPTS

• **Applied computing** → **Computer games**; • **Theory of computation** → *Representations of games and their complexity*.

KEYWORDS

Game AI, Match play, Match strategy, Tavli, Backgammon, Game complexity, First player advantage

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1 INTRODUCTION

Games, traditionally seen as a measure of human cognitive abilities, have also been one of the key battleground for testing AI algorithms against human intelligence. With the successes of AlphaGo [11], AlphaZero [10] and their derivatives [9], games is the first domain where AI has surpassed human experts in classical board games and other video games [1, 13, 15]. However, these advances assume that the games are played independently, focusing on accumulating as many points as possible in individual games.

On the other hand, human game competitions rarely have this limited goal. Most of the times human competition in games uses individual games as a means to a next level goal. The most common competitions used by humans is head-to-head matches and tournaments. The goal in a head-to-head match is to win the match and in tournaments is usually to achieve as higher standing as possible especially when prizes are awarded according to placement.

In order to achieve these competition goals humans recognize that sometimes they have to deviate from the usual strategy. For example in chess matches of fixed length when a player is behind in match she knows that she needs to take greater risks in the following games in order to win the match.

In another example, in poker tournaments players are awarded money according to placement and only a small percentage gets any money at all. At a certain point in the tournament when the number of players eliminated gets near the last unpaid player (also called the bubble) their play gets more defensive in order to not get eliminated so close to a payment.

In both examples players adapt their strategy according to the match/tournament situation because there exists a higher goal that supersedes the usual mantra "play the best individual game that you can".

One of the first examples highlighting the importance of being aware of the match situation and acting accordingly was the 100-game 1998 AAAI Hall of Champions backgammon exhibition match between World Cup Champion Malcolm Davis and TD-Gammon version 3.1 [12]. After a blunder by TD-Gammon that resulted in Davis taking the lead late in the match, Davis played conservatively to protect his lead, while TD-Gammon, unequipped with any match strategy, lacked the required urgency and ultimately lost the match.

In this paper we will address the problem of finding a good strategy to play the backgammon variant of Tavli, popular in Greece and neighboring countries, in a match setting. First, we calculate the winning probabilities of the player who plays first in any game of a Tavli match. We use these probabilites to construct effective Tavli AI strategies, and to gain valuable insights on the Tavli game.

Second, we evaluate the created Tavli AI match strategy against existing AI strategies that do not take account the match setting. We find that the Tavli AI match strategy wins by a small, but statistical significant margin, and we also discuss potential reasons for these results.

Lastly, the resulting Tavli match strategy was evaluated against human opponents via an Android app published on the Android app store. We find that the constructed AI match strategies significantly outperform humans, and we compare these results with other match types, concluding that a Tavli match is one of the most difficult match settings for humans.

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2 RELATED WORK

Backgammon is a popular board game typically played in two formats: "money play" or "match play." In "money play", each point counts equally and every game is independent, whereas in a "match", players compete in a series of games, with victory awarded to the first player to reach (or exceed) a predetermined number of points. Multiple AI software exist for backgammon [2, 4–6] and human players used extensively these tools to improve their play both for money and match play.

2.1 The Tavli variant

Tavli is the most popular way of playing backgammon in Greece. It comprises three distinct backgammon variants played without the doubling cube in a cycle: Portes, Plakoto and Fevga until one player reaches a pre-specified number of points (usually 5 or 7). Thus, Tavli is only played in a match setting. The rules of the individual games can be found in [8]. The order of the games is usually Portes-Plakoto-Fevga, but in some Greek regions, another order is used: Fevga-Plakoto-Portes. We will examine both of these orders of play. The doubling cube, typically used in backgammon, is not utilized in Tavli matches.

Compared to a standard backgammon match, a Tavli match has several unique challenges:

- `3 different games`: Players need to know how to play well three different backgammon variants. With the exception of the final stage of the game called "bearing-off" which is the same in all three, the style of play is completely different in each variant. In contrast, backgammon matches are only played with one game.
- `Not fully studied`: While Portes is almost similar to backgammon, the other two games are not studied as extensively as backgammon. There are very few books for helping human players, and Tavli software were made available only recently, compared to standard backgammon.
- `First player is always the previous winning player`: This rule is unique to the Tavli variant. In a typical backgammon match, the first player of a game is always random.
- `Different match strategy`: In a backgammon match (or any match with a single game type) the strategy of the player changes only by the score of the match. In a Tavli match where game types alternate in a cycle, a successful match strategy should also consider the next game type in the order. For example, losing a Plakoto game in a Portes-Plakoto-Fevga order has more severe consequences, since the first player of a Fevga game is known to have great advantage compared to the second player [8].

2.2 Match strategy in backgammon games

The prevalent method of obtaining a good match strategy in all backgammon bots is through the use of Match Equity Table (MET). A MET is a matrix that shows the probability of winning a match from various scores. As an example, we show the first entries of one of the first MET created by human experts in Table 1.

METs are laid out according to the number of points each player still needs to win the match also named as "away points". METs are used to transform the values estimated by the AI for winning the game (game equities) to values estimated for winning the match (match equities), also known as Match Winning Chances (MWC), according to the following equation:

Table 1: Example of a Backgammon MET created by Kit Woolsey [14] (shown up to 5-away points for brevity)

	1	2	3	4	5
1	50	70	75	83	85
2	30	50	60	68	75
3	25	40	50	59	66
4	17	32	41	50	58
5	15	25	34	42	50

$$MWC = ws * M_{a-1,b} + wd * M_{a-2,b} + ls * M_{a,b-1} + ld * M_{a,b-2} \quad (1)$$

where ws, wd, ls and ld are the outcome estimations of our estimator (usually a neural network), a the points needed for player A to win the match, b the the points needed for player B to win the match and $M_{i,j}$ is the i,j element of the MET matrix containing the probability of player A winning a match when the match score is $i - away, j - away$ before the start of the game. When one index goes outside the matrix bounds, this indicates a match end, and the value of corresponding element of M is replaced with 1 (when used with the ws, wd estimates) or with 0 (when used with the ls, ld estimates). Note that our estimator must be able to estimate the full probability distribution of the outcomes and not just a single value.

While there have been several methods to calculate a MET and a few other approaches to estimate the match equity, most of these attempts were applied in standard backgammon (with cube). Strategies for match play in matches of the same game type have been also investigated for Portes, Plakoto and Fevga in [8]. To the best of our knowledge, there have been no other attempts to create METs, or to construct effective strategies for a Tavli match.

3 CONSTRUCTING AI STRATEGIES FOR TAVLI MATCH PLAY

As with other backgammon software, we tackle the problem of playing correctly in a Tavli match via constructing METs for the Tavli game.

3.1 Tavli METs

We construct three Tavli METs, one for each game type (Portes, Plakoto, Fevga). The iterative algorithm that we used is presented in Algorithm 1. The values stored in the elements of these METs correspond to the probability of the first player winning the match. Therefore we maintain that player A (the player on the rows) always plays first when the game starts, and player B (the player on columns) always plays second.

The resulting METs along with details of their construction are shown in appendix B. Compared to standard backgammon METs the Tavli METs have the following differences:

- We need 3 different METs one for each game in the game cycle.
- The element values should be calculated from the MET belonging to the next game type in the cycle. A backgammon

MET could be constructed by referencing itself since the next game is always of the same game type.

- For a Tavli match score at i-away, j-away the value of the MET element M[i,j] indicates the probability of the first player winning the Tavli match. Consequently, the winning chance of the second player to win from the same match score is 1 - M[i,j]. In contrast, in a typical backgammon match the winning chance of the first player and the second player is the same since determining who plays first is random.

Algorithm 1 Tavli METs Generation

Inputs: l is the length of the match, $PortesP$, $PlakotoP$, $FevgaP$ are the outcome probabilities of first player for each game
Initialize:
l X l Matrices $PortesMET$, $PlakotoMET$, $FevgaMET$
Vector $gPs \leftarrow [PortesP, PlakotoP, FevgaP]$
Vector $mets \leftarrow [PortesMET, PlakotoMET, FevgaMET]$
Vector $next \leftarrow [1, 2, 0]$ ▷ Index of next game
for $i = 0, \ldots, l-1$ **do**
  for $j = 0, \ldots, i$ **do**
    for $g = 0, \ldots, 2$ **do**
      updateMet($mets[g], gPs[g], mets[next[g]], i, j$);
return $mets$
function UPDATEMET($met, gP, nextMet, i, j$)
  $met[i, j] \leftarrow mwc(nextMet, gP, i, j)$
  if $i \neq j$ **then**
    $met[j, i] \leftarrow mwc(nextMet, gP, j, i)$
function MWC($nextMet, gP, i, j$)
  $ws \leftarrow gP.WS * nextMet[i-1, j]$ ▷ 1 if $i - 1 < 0$
  $wd \leftarrow gP.WD * nextMet[i-2, j]$ ▷ 1 if $i - 2 < 0$
  $ls \leftarrow gP.LS * (1 - nextMet[j-1, i])$ ▷ 0 if $j - 1 < 0$
  $ld \leftarrow gP.LD * (1 - nextMet[j-2, i])$ ▷ 0 if $j - 1 < 0$
  return $ws + wd + ls + ld$

3.2 Tavli Match Strategy

With the METs for all three game types of a Tavli match calculated, we can now estimate the Tavli match winning chance (TWC) of each candidate move using the following equation:

$$TWC = ws*N_{a-1,b} + wd*N_{a-2,b} + ls*(1-N_{b-1,a}) + ld*(1-N_{b-2,a}) \quad (2)$$

where ws, wd, ls and ld are the outcome estimations of our neural network estimator, a the points needed for player A to win the match, b the points needed for player B to win the match and $N_{i,j}$ is the i,j MET element of the next game in the cycle containing the probability of player A winning the Tavli match when the match score is $i - away$, $j - away$ before the start of the game. Similarly with MWC in 1, when one index goes outside the matrix bounds, this indicates a match end, and the value of corresponding element of N is replaced with 1 (when used with the ws, wd estimates) or with 0 (when used with the ls, ld estimates).

In contrast with MWC, in the event of losing the match (ls, ld), instead of taking the value in index [i,j], we take the value with indexes reversed (1 - N[j,i]). This happens because the Tavli METs contain the values of the first player, and in the event of losing, the first player of the next game becomes the opponent. Therefore, we have to take the element from the opponent's point of view.

Table 2: First player's match winning chance (MWC) in all possible Tavli game order configurations, with and without doubles in the first roll of the match.

Game Order	1st roll w double 5-point	1st roll w double 7-point	1st roll w/o double 5-point	1st roll w/o double 7-point
Portes-Plakoto-Fevga	**0.5089**	**0.5073**	**0.5047**	**0.5039**
Plakoto-Fevga-Portes	0.5124	0.5101	0.5085	0.5070
Fevga-Portes-Plakoto	0.5235	0.5194	0.5217	0.5179
Fevga-Plakoto-Portes	0.5240	0.5198	0.5221	0.5182
Plakoto-Portes-Fevga	0.5116	0.5095	0.5080	0.5065
Portes-Fevga-Plakoto	0.5097	0.5080	0.5051	0.5042

3.3 Game order in a Tavli match

The prevalent game order of a Tavli match in Greece is Portes-Plakoto-Fevga, but in some regions another less frequent order is used: Fevga, Plakoto, Portes. In order to examine which order is the most fair to the player playing second, we have constructed also the Tavli METs of this alternate order (Tables 9, 10, 11). With the Tavli METs calculated, we can extract the winning chances of the first player before the start of the match by taking the corresponding values from the matrix diagonal. For example for the "classical" order Portes-Plakoto-Fevga, the winning chance of the first player in a 7-point Tavli match can be extracted from the [7,7] element of Table 6.

The winning chance of the player that plays first in a Tavli match for the most prevalent match lengths (5-point, 7-point), for all possible game orders is summarized in Table 2. We do not have to construct METs for other orders since the METs calculated for the Portes-Plakoto-Fevga order can also be used for the Plakoto-Fevga-Portes order, etc.

We also examined another equally popular variation that does not allow the first roll of the match to be a double. To construct the winning chance in this case, we changed only the outcome distribution of the current game by removing the double rolls for the scores in question (5a-5a and 7a-7a), while keeping the next game's value from the same MET. Unsurprisingly, in all variations examined, the win rate of the first player was lower when no double was allowed because it is already established that a double roll increases the first player's winning chance in all games [8].

As we can see from Table 2, the order that loses less for the second player is the order Portes-Plakoto-Fevga order in all configurations examined. On the other hand, the two orders starting with Fevga are the most unfair to the second player. This is not so surprising since the game of Fevga is known to give great advantage to the first player, and when this is the first game of the match this is propagated to the final result of the match.

4 EVALUATION VS OTHER AI STRATEGIES

The resulting Tavli match AI strategy was first evaluated against the individual game AI strategies in 100000 matches of lengths 1 to 7. We used our best neural network evaluators (named PortesACG13,

Table 3: Win percentages (out of 100000 matches) of Tavli match AI strategy versus individual game AI strategies. The last column shows the percentage of moves played differently by the match AI.

Match Length	Wins (%)	Diff. Moves
7	50.31%	2.35%
6	50.44%	2.58%
5	50.46%	2.94%
4	50.59%	3.52%
3	50.61%	4.77%
2	50.84%	7.01%
1	50.50%	12.75%

Plakoto-5, Fevga-6)[7] for estimating the outcome probabilities of each move. The Tavli AI match strategy always selected the best move that maximized the TWC, while the game AI strategy always selects the move that maximizes the value of the individual game played. Both strategies used did not use any search (i.e. 1-ply setting) and utilized the same NN evaluators. In order to have an equal comparison, the first player alternated at the start of each match. We also alternated the starting game type at the beginning of the match because we didn't want a specific starting game type to have much influence in the final score. This is especially true at lower match lengths, since these match scores can be reached from higher lengths also. All win rates have a 95% confidence interval (CI) of 0.3%. Additionally, for each non-trivial move decision (move count > 1), we compared the move selected with the move the game AI strategy would have selected and we kept a counter when these moves differed.

The results are shown in Table 3. From the results, we can see that the match AI strategy has a small but statistical significant advantage versus the game AI strategy. This advantage increases slowly as the match length decreases, and we can see why this happens in the last column that keeps track of the different moves: As the match length increases, the difference between the moves played by the match strategy and the game strategy decreases. In other words, the match strategy becomes more important as the match comes to its end.

5 EVALUATION VS HUMAN PLAYERS

To compare the resulting Tavli AI strategy with that of humans, we published an Android application to the Android Play Store[1] where humans can play against the different AI strategies and we logged the results.

5.1 Data collection and filtering

During the period of 3.5 years (from April 2020 up to October 2023), we measured Tavli matches (of lengths 5,7) and individual game type matches (of lengths 1,3,5,7). We used the single type match AI strategies created in [8] for the single type matches and the Tavli match AI strategy created in this paper for the Tavli matches. All Tavli matches where conducted with the Portes-Plakoto-Fevga order and a double roll was allowed in the first roll of the match. As in the AI evaluation, we used our best neural network evaluators for estimating the outcome probabilities of each move. The user had the option of three difficulty levels: *Easy*, *Normal* and *Expert*. We collected data from the *Normal* and *Expert* levels which correspond to 1-ply and 2-ply search respectively.

During the match, the user has the option to abandon the match after the end of each game. We did not include abandoned matches in the results; only completed matches were included. The number of abandoned matches was significant: at least as many as the completed matches in the 5-point lengths, and more than double the amount of completed matches in the 7-point length. For the same reason, we excluded 1-point matches, as resigning immediately ended the match.

Lastly, we observed a behavior pattern in a small number of players (15) who abandoned most of their matches early and only completed a small percentage of the matches they started. We excluded all data from these players.

[1] https://play.google.com/store/apps/details?id=gr.nikpapa.palamedes

Table 4: Human performance against match AI strategies

Match Type	Match Length	Count	Wins (%) ± CI
Portes	5	3908	23.93% ± 1.34
Plakoto	5	1748	30.15% ± 2.15
Fevga	5	2836	14.99% ± 1.31
Tavli	5	9154	17.52% ± 0.77
Tavli	7	13115	16.30% ± 0.63

5.2 Human win rates

In Table 4 we show the combined performance of humans against various match AI strategies. We did not include the results of individual game matches of 7 points because the match count was significantly less than the ones from the 5-point length. For brevity, the results of the *Normal* and *Expert* difficulty levels were combined.

The results show that all AI strategies significantly outperform humans, with Fevga matches exhibiting the worst performance for humans closely followed by Tavli matches. This is not surprising, we have already outlined the challenges of a Tavli match in Section 2.

5.3 ELO Ratings

The above analysis shows the strength of the AI strategies against the combined human players who played our application, but it does not exclude the possibility of some humans being better than the AI.

To accurately compare the playing strength of our approach, we also computed the Elo rating [3] for all AI strategies and human players. The Elo rating system is a method for calculating the relative skill levels of players, which has been widely adopted in competitive games and sports. The difference in the ratings between two players serves as a predictor of the outcome of a match. Using the collected data, we calculated all the Elo ratings, and show the Elo evolution of the AI strategies in Figure 1. We also show the mean Elo of the AI strategies and the max Elo reached by the

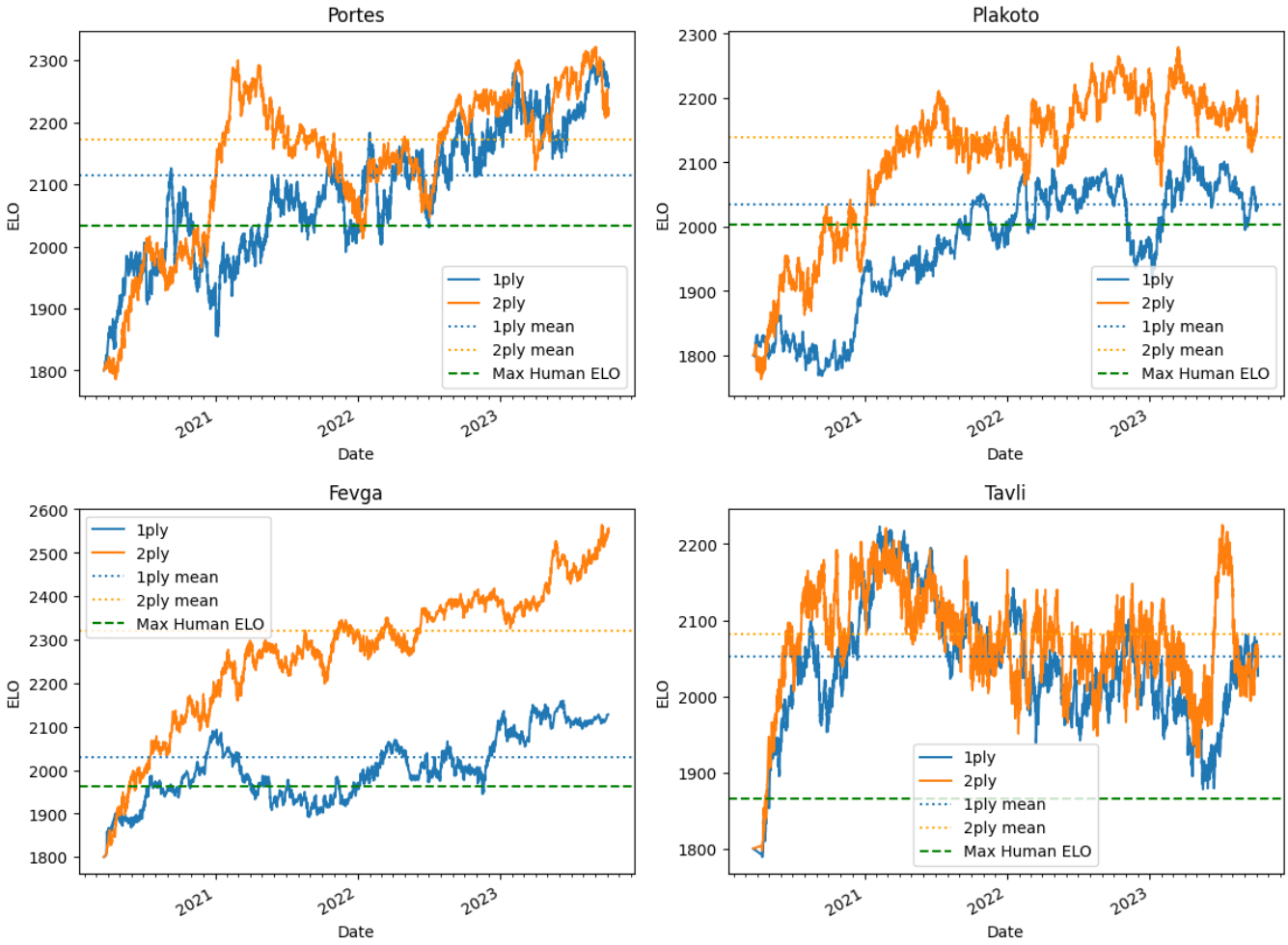


Figure 1: Online evolution of Elo ratings for various match AI strategies using the data collected from our Android application. Blue lines indicate 1-ply search and orange lines 2-ply search. All diagrams show that the mean Elo rating of AIs exceed that of the best humans (green lines).

best human. More details for the Elo calculations are shown in appendix A

The calculated Elo ratings for the AI strategies and human players show that no human player who played in our application managed to surpass the ratings of our AI players in any match setting examined. The performance gap is most significant in the Fevga and Tavli match settings.

6 CONCLUSION

We presented a way to construct Tavli match strategies through the creation of Tavli METs. These METs contain the winning chances for the first player to move for every match situation and every game type found in a Tavli match. To the best of our knowledge, there have been no other attempts to create Tavli METs or to develop strategies that consider the intricacies of the Tavli match setting. The effectiveness of our approach was tested versus other AI strategies, where we found a small but statistically significant improvement, and versus human opponents, where we found that the resulting AI strategy significantly outperforms humans.

Another useful insight derived from the Tavli METs was the examination and comparison of all game type orders in Tavli matches. We found that the order Portes-Plakoto-Fevga is the most fair to the second player because it gives the lowest advantage to the first player.

Even though our methods and experiments were applied to the backgammon variant of Tavli, we believe that they can be also applied to other games that are played in a match setting. We also plan to apply our methods to other competition types such as swiss tournaments, round-robin tournaments, etc.

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A ELO CALCULATION DETAILS

The typical equations used for calculating Elo ratings are:

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}} \tag{3}$$

where E_A is the probability of player A winning, R_A, R_B are the current ratings of players A and B, and

$$R'_A = R_A + K(S_A - E_A) \tag{4}$$

where R'_A is the new rating calculated for player A, K is a scaling factor and aS_A is the points won or lost by player A. K is typically larger for new players in order to quickly change their rating.

In backgammon matches, a modification has been proposed where equations 3 and 4 are adjusted to take into account the length of the match:

$$E_A = \frac{1}{1 + 10^{\sqrt{N}(R_B - R_A)/2000}} \tag{5}$$

$$R'_A = R_A + K\sqrt{N}(S_A - E_A) \tag{6}$$

where N is the length of the match and K scaling factor is defined as:

$$K = 5 * max(1, 5 - \frac{XP_A}{100}) \tag{7}$$

Table 5: First player outcome probabilities

Game	WS(+1)	WD(+2)	LS(-1)	LD(-2)
Portes (all rolls)	0.3818	0.1462	0.3487	0.1233
Plakoto (all rolls)	0.299	0.2277	0.2962	0.1771
Fevga (all rolls)	0.4731	0.1022	0.3842	0.0405
Portes (w/o doubles)	0.3745	0.1403	0.3570	0.1282
Plakoto (w/o doubles)	0.3031	0.2154	0.3009	0.1806
Fevga (w/o doubles)	0.4730	0.0969	0.3889	0.0412

where XP_A is the experience accumulated by player A, calculated as the sum of all the match lengths played so far.

Since this modification was specifically designed for backgammon, it may not be entirely accurate for Tavli and other game types. In the absence of alternative rating methods for backgammon variants, we applied the same equations (5, 6, 7) to all game types. Assessing the suitability of this rating system for these variants is left for future work.

In backgammon, the starting rating for humans is typically set to 1500, and we maintained this for our human players. For the AI strategies, we raised the starting Elo to 1800 to avoid penalizing the human opponents early in the calculations when the AI has a very low rating.

As we can we see in Figure 1 the ratings of the AI strategies have a lot of fluctuations, due to luck and varying opposition strength. For this reason, we believe the best rating assigned to the AI strategies is not the final rating achieved, but the mean rating after a sufficient number of matches have been played (we use the number 1000 in this paper). Human players, unlike AIs, do not have a fixed performance and may become better (or worse) over time. Therefore, for human players, we select the maximum Elo rating achieved, which signifies their peak strength.

B TAVLI METS

We present here the Tavli METs generated by Algorithm 1. The outcome probabilities of each game (PortesP, PlakotoP, FevgaP) were previously calculated in [8]. We present the values used in this paper in Table 5 where WS is the probability of winning a single game, WD the probability of winning a double game, LS the probability of losing a single game and LD the probability of losing a double game. The first three rows show the distribution when doubles are allowed in the first roll, while the next three rows show the distribution when doubles are not allowed in the first roll of a Tavli match. We used the first three rows to construct the Tavli METs and the last three rows to calculate the first player's winning chance when a double roll is not allowed in the first game of a Tavli match.

We assume that this probability distribution is constant throughout the match and in every match situation. While this may not be accurate in practice, we believe that any differences in the outcome probabilities for a few match scores should be small and not influential enough to change the results. We leave a more fine-grained outcome distribution calculation for every match situation for future work.

The METs of the alternate game type order (Fevga-Plakoto-Portes) are also included. The only change in Algorithm 1 needed for this order is to replace line 6 (Index of next game) with $next \leftarrow [2, 0, 1]$.

Table 6: Portes MET for the Portes-Plakoto-Fevga Tavli order

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	0.528	0.693	0.796	0.868	0.920	0.950	0.967	0.979	0.987	0.992	0.995	0.997	0.998
2	0.347	0.516	0.657	0.753	0.827	0.879	0.917	0.943	0.961	0.974	0.982	0.988	0.992
3	0.230	0.368	0.512	0.627	0.720	0.793	0.848	0.890	0.921	0.944	0.960	0.972	0.980
4	0.149	0.267	0.395	0.510	0.612	0.698	0.768	0.824	0.868	0.902	0.928	0.947	0.961
5	0.090	0.188	0.298	0.407	0.509	0.601	0.681	0.748	0.804	0.849	0.885	0.913	0.934
6	0.057	0.132	0.221	0.318	0.415	0.508	0.593	0.668	0.732	0.787	0.832	0.869	0.899
7	0.037	0.091	0.163	0.245	0.334	0.422	0.507	0.586	0.657	0.719	0.773	0.818	0.856
8	0.024	0.062	0.118	0.187	0.264	0.346	0.428	0.507	0.581	0.648	0.708	0.761	0.806
9	0.015	0.043	0.085	0.141	0.207	0.279	0.356	0.432	0.506	0.576	0.641	0.699	0.750
10	0.009	0.029	0.061	0.105	0.160	0.223	0.292	0.364	0.436	0.506	0.573	0.634	0.690
11	0.006	0.020	0.043	0.078	0.122	0.176	0.237	0.302	0.371	0.439	0.506	0.569	0.629
12	0.004	0.013	0.030	0.057	0.093	0.138	0.190	0.249	0.312	0.377	0.442	0.505	0.566
13	0.002	0.009	0.021	0.041	0.070	0.107	0.151	0.203	0.259	0.320	0.382	0.444	0.505

Table 7: Plakoto MET for the Portes-Plakoto-Fevga Tavli order

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	0.527	0.652	0.794	0.873	0.919	0.947	0.966	0.979	0.987	0.992	0.995	0.997	0.998
2	0.400	0.526	0.657	0.756	0.828	0.881	0.918	0.943	0.961	0.974	0.982	0.988	0.992
3	0.236	0.382	0.518	0.632	0.723	0.795	0.850	0.891	0.922	0.944	0.960	0.972	0.980
4	0.145	0.273	0.398	0.514	0.616	0.700	0.770	0.825	0.869	0.902	0.928	0.947	0.962
5	0.093	0.192	0.301	0.410	0.512	0.604	0.683	0.750	0.805	0.850	0.885	0.913	0.935
6	0.061	0.133	0.224	0.322	0.419	0.511	0.595	0.670	0.734	0.789	0.834	0.870	0.900
7	0.039	0.092	0.165	0.248	0.337	0.425	0.510	0.589	0.659	0.721	0.774	0.819	0.857
8	0.024	0.064	0.120	0.189	0.267	0.349	0.430	0.509	0.583	0.650	0.710	0.762	0.807
9	0.015	0.044	0.087	0.143	0.209	0.282	0.358	0.435	0.509	0.579	0.643	0.700	0.751
10	0.010	0.030	0.062	0.107	0.162	0.225	0.294	0.366	0.438	0.508	0.575	0.636	0.692
11	0.006	0.020	0.044	0.079	0.124	0.178	0.239	0.305	0.373	0.441	0.508	0.571	0.631
12	0.004	0.014	0.031	0.058	0.094	0.139	0.192	0.251	0.314	0.379	0.444	0.508	0.568
13	0.002	0.009	0.022	0.042	0.071	0.108	0.153	0.204	0.261	0.322	0.384	0.446	0.507

Table 8: Fevga MET for the Portes-Plakoto-Fevga Tavli order

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	0.575	0.757	0.845	0.898	0.934	0.959	0.975	0.984	0.989	0.993	0.996	0.997	0.998
2	0.352	0.548	0.677	0.775	0.845	0.893	0.926	0.950	0.966	0.977	0.985	0.990	0.993
3	0.218	0.394	0.532	0.647	0.738	0.807	0.860	0.900	0.928	0.949	0.964	0.975	0.983
4	0.144	0.278	0.409	0.527	0.628	0.713	0.781	0.835	0.877	0.909	0.933	0.951	0.965
5	0.094	0.195	0.309	0.420	0.524	0.616	0.694	0.760	0.814	0.858	0.892	0.919	0.939
6	0.058	0.136	0.230	0.329	0.428	0.521	0.606	0.680	0.744	0.797	0.841	0.877	0.905
7	0.036	0.094	0.168	0.254	0.344	0.434	0.519	0.598	0.669	0.730	0.782	0.826	0.863
8	0.023	0.064	0.122	0.193	0.273	0.356	0.438	0.518	0.592	0.659	0.718	0.770	0.814
9	0.015	0.044	0.088	0.145	0.213	0.287	0.365	0.442	0.517	0.587	0.651	0.708	0.759
10	0.009	0.030	0.063	0.108	0.165	0.230	0.300	0.373	0.445	0.516	0.582	0.644	0.699
11	0.006	0.020	0.045	0.080	0.126	0.181	0.243	0.310	0.379	0.448	0.515	0.579	0.638
12	0.004	0.014	0.031	0.059	0.096	0.142	0.196	0.255	0.319	0.385	0.450	0.514	0.575
13	0.002	0.009	0.022	0.043	0.072	0.110	0.156	0.208	0.266	0.327	0.390	0.453	0.514

Table 9: Fevga MET for the Fevga-Plakoto-Portes Tavli order

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	0.575	0.757	0.831	0.896	0.934	0.959	0.973	0.983	0.989	0.993	0.996	0.997	0.998
2	0.351	0.546	0.676	0.777	0.844	0.892	0.926	0.950	0.966	0.977	0.985	0.990	0.993
3	0.236	0.394	0.533	0.647	0.738	0.808	0.861	0.900	0.928	0.949	0.964	0.975	0.983
4	0.148	0.277	0.410	0.528	0.629	0.713	0.781	0.835	0.877	0.909	0.933	0.952	0.965
5	0.094	0.196	0.309	0.421	0.524	0.616	0.695	0.761	0.815	0.858	0.892	0.919	0.939
6	0.059	0.137	0.229	0.329	0.428	0.522	0.606	0.681	0.744	0.798	0.841	0.877	0.905
7	0.038	0.094	0.168	0.254	0.344	0.434	0.520	0.599	0.669	0.730	0.783	0.827	0.863
8	0.024	0.064	0.122	0.193	0.273	0.356	0.439	0.518	0.592	0.659	0.719	0.770	0.814
9	0.015	0.044	0.088	0.145	0.213	0.288	0.365	0.442	0.517	0.587	0.651	0.709	0.759
10	0.010	0.030	0.063	0.108	0.165	0.230	0.300	0.373	0.446	0.516	0.583	0.644	0.700
11	0.006	0.020	0.045	0.080	0.126	0.181	0.243	0.310	0.379	0.448	0.515	0.579	0.638
12	0.004	0.014	0.031	0.059	0.096	0.142	0.195	0.255	0.319	0.385	0.451	0.515	0.576
13	0.002	0.009	0.022	0.043	0.072	0.110	0.155	0.208	0.266	0.327	0.390	0.453	0.514

Table 10: Plakoto MET for the Fevga-Plakoto-Portes Tavli order

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	0.527	0.667	0.798	0.871	0.919	0.948	0.967	0.979	0.987	0.992	0.995	0.997	0.998
2	0.386	0.526	0.668	0.758	0.830	0.882	0.919	0.944	0.962	0.974	0.983	0.988	0.992
3	0.230	0.367	0.516	0.631	0.723	0.795	0.850	0.892	0.922	0.945	0.961	0.972	0.981
4	0.149	0.269	0.397	0.513	0.615	0.701	0.770	0.826	0.870	0.903	0.929	0.948	0.962
5	0.092	0.190	0.299	0.409	0.512	0.604	0.684	0.751	0.806	0.851	0.886	0.914	0.935
6	0.059	0.132	0.222	0.320	0.418	0.510	0.595	0.670	0.735	0.789	0.834	0.871	0.900
7	0.038	0.091	0.164	0.247	0.335	0.424	0.510	0.588	0.659	0.722	0.775	0.820	0.857
8	0.024	0.063	0.119	0.188	0.265	0.347	0.429	0.509	0.583	0.650	0.710	0.763	0.807
9	0.015	0.043	0.086	0.141	0.207	0.281	0.357	0.434	0.508	0.578	0.643	0.701	0.752
10	0.010	0.029	0.061	0.105	0.160	0.224	0.293	0.365	0.437	0.508	0.574	0.636	0.692
11	0.006	0.020	0.043	0.078	0.123	0.177	0.238	0.304	0.372	0.440	0.507	0.571	0.631
12	0.004	0.013	0.031	0.057	0.093	0.138	0.191	0.250	0.313	0.378	0.443	0.507	0.568
13	0.002	0.009	0.021	0.041	0.070	0.107	0.152	0.203	0.260	0.321	0.383	0.445	0.507

Table 11: Portes MET for the Fevga-Plakoto-Portes Tavli order

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	0.528	0.676	0.807	0.874	0.919	0.949	0.968	0.980	0.987	0.992	0.995	0.997	0.998
2	0.366	0.520	0.652	0.756	0.829	0.882	0.918	0.944	0.962	0.974	0.983	0.988	0.992
3	0.218	0.378	0.513	0.630	0.722	0.794	0.850	0.891	0.922	0.944	0.961	0.972	0.981
4	0.142	0.267	0.393	0.511	0.613	0.699	0.769	0.825	0.869	0.903	0.928	0.948	0.962
5	0.091	0.187	0.297	0.407	0.510	0.602	0.682	0.750	0.805	0.850	0.886	0.914	0.935
6	0.057	0.130	0.221	0.318	0.416	0.509	0.594	0.669	0.734	0.788	0.834	0.870	0.900
7	0.036	0.090	0.162	0.245	0.333	0.423	0.508	0.587	0.658	0.721	0.774	0.819	0.857
8	0.023	0.062	0.117	0.186	0.264	0.346	0.428	0.507	0.582	0.649	0.709	0.762	0.807
9	0.015	0.042	0.084	0.140	0.206	0.279	0.356	0.432	0.507	0.577	0.642	0.700	0.751
10	0.009	0.029	0.060	0.104	0.159	0.222	0.292	0.364	0.436	0.507	0.573	0.635	0.691
11	0.006	0.019	0.043	0.077	0.122	0.175	0.236	0.302	0.371	0.439	0.506	0.570	0.630
12	0.004	0.013	0.030	0.056	0.092	0.137	0.190	0.248	0.311	0.377	0.442	0.506	0.567
13	0.002	0.009	0.021	0.041	0.069	0.106	0.151	0.202	0.259	0.319	0.382	0.444	0.506